



MECH MAINT- OFF SITE Department
SIMHADRI

Ref:SMTTPP/MECH MAINT-/2022-23/NTPCSIMH/172479

Date:

10/06/2022(12:01:51)

Subject:APPROVAL OF LMI- GUIDELINES FOR COOLING TOWER
MAINTENANCE

LMI GUIDELINES FOR COOLING TOWER MAINTENANCE (LMI/OGN/OPS/MECH/024) Issue-3 was prepared based upon OGN OGN/OPS/MECH/024/REV 0, MAR'05.

Issue-4 of the said LMI is being circulated (no change) for approval.

Submitted for approval.

Submitted by:

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CORRECTED LMI PLEASE APPROVED

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Revision may please be accorded approval.

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कृपया अनुमोदन प्रदान किया जाए।

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LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/024

ISSUE NO. 01; REVISION NO. 2.0; DATE: JUN'2022

GUIDELINES FOR COOLING TOWER
MAINTENANCE

APPROVED FOR IMPLEMENTATION

HOP(SIMHADRI)

Date-----

GUIDELINES FOR COOLING TOWER MAINTENANCE

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	GUIDELINES FOR COOLING TOWER MAINTENANCE
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/024/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV.-2.00; JUN'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/024/REV 0, MAR'05</u>
<u>4.PREPARED BY</u>	NAME: J.P. SINGH Sr. Manager (MM/OS) SIGNATURE-----
<u>5. CHECKED BY</u>	NAME: D.Sri Rama sastray DGM (MM-OS) SIGNATURE-----
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<u>7. APPROVED BY</u>	NAME: Rajesh Kumar Jha GM(O&M) SIGNATURE:-----
<u>8.DATE OF ISSUE</u>	10-06-2022
<u>9.INDEX LIST</u>	ENCLOSED

GUIDELINES FOR COOLING TOWER MAINTENANCE

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GUIDELINES FOR COOLING TOWER MAINTENANCE

12.0

COMPLIANCE RECORD

11

1.0 INTRODUCTION

Cooling towers play an important role in ensuring the thermal efficiency of a generating station. Thermal efficiency of a steam turbine is dependent on the temperature of the exhaust steam from the LP Turbine into the condenser. Heat from the steam side of the condenser is transferred to the circulating water. This heat is in turn transferred to the atmosphere in Cooling tower. If the cooling tower does not perform satisfactorily, the cold water temperature of the cooling water would go up. This in turn would affect the condensing temperature of steam and back pressure in the condenser. This would result in an increased heat rate and subsequent increase in fuel input. Regular checks and maintenance would result in the efficiency of the cooling tower.

This LMI is based on the Operation Guidance note OGN/OPS/MECH/024/REV 0, MAR’05

2.0 SUPERSEDED DOCUMENT

Nil

3.0 INSPECTION AND MAINTENANCE OF COOLING TOWER

SL NO	COMPONENT	CHECKS	FREQUENCY	RESPONSIBILITY
1	Concrete Structure	- Inspection of the entire RCC Structure of cooling Tower. Check if any reinforced steel is exposed to atmosphere and repair. - Anti corrosive paint to be applied on structure and cold water basin	QUARTERLY Once in 10 years or during Capital Overhaul	O&M-CIVIL
2	Stairs	- Check for evidence of steel corrosion. - Check for loose treads, handrails or stringers. - Check that all hardware is in good condition and tight.	QUARTERLY	O&M-CIVIL/MM-OS
3	Ladders	- Check the general condition of the material - Check that all hardware is in good condition and all connections between the tower and ladder are secure.	QUARTERLY	O&M-CIVIL

GUIDELINES FOR COOLING TOWER MAINTENANCE

SL NO	COMPONENT	CHECKS	FREQUENCY	RESPONSIBILITY
4	Walkways	- Check the general condition of the material. - Check for loose treads or handrails. - Check that all hardware is in good condition and tight. - Particular attention should be paid to any damage or deterioration that may pose a potential hazard to personnel.	QUARTERLY	O&M-CIVIL
5	Cooling Tower Surrounding	The cooling tower surroundings up to at least 30m should be free from trees or bushes etc. The area between two towers should also be totally clean. The area must be covered with small grass or paved with ash bricks. Fresh air entering CT picks up moisture from trees and its cooling capability decreases and in summer the dry leaves fall on ground and are carried away with air to the CT which chokes nozzles and also chokes condenser if they pass through nozzles and fills.	Regularly (As per requirement)	O&M-CIVIL
6	Cold Water Basin	- Check for excessive build-up of sludge or debris. Clean as required. - Check the general condition of the basin material. - Check the basin outlet grill to condenser for choking - Check for any overflow of basin. - Cleaning of basin to be done minimum once during unit O/H	During unit OH MONTHLY During unit OH	O&M-CIVIL/ MM-OS
7	Distribution Ducts	- Check the general condition of the material. Check for leaks and holes. Repair where required. - Check for algae & debris and clean.	During Overhaul	O&M-CIVIL/ MM-OS

GUIDELINES FOR COOLING TOWER MAINTENANCE

SL NO	COMPONENT	CHECKS	FREQUENCY	RESPONSIBILITY
8	Hot Water Distribution Piping	- Examine all supports. - Check for leaks at all connections and Check the general condition of all piping looking for signs of deterioration or breaking of asbestos cement pipes/ PVC pipes and end caps.	During Overhaul	O&M-CIVIL/ MM-OS
9	Spray Nozzles	- Check for missing or damage. - Check for clogged/ blocked nozzles. - Check that all nozzle components are in place and working properly (i.e. splashers)	During Overhaul.	MM-OS
10	Splash Fill	- Check the condition of the splash bars. Check for sagging, broken, missing, heavily-scaled pieces. If splash bar is broken or not in position then new bars are to be put. - Check the support grids and Check the support members for deterioration and damage.	During Overhaul	MM-OS
11	Drift Eliminators	Check for cleanliness so that all air passages are clear.	During Overhaul	MM-OS
12	Lightning arrestor	Check that the connections to the lightning arrestor are secure and the connecting wires are clamped and routed along the circumference of the CT and earthed properly. Earth resistance measurements to be recorded.	YEARLY	EMD
13	Aviation Lighting	Aviation lights in NDCT along with fittings shall be checked at the top of the CT and replaced if necessary.	YEARLY	EMD

GUIDELINES FOR COOLING TOWER MAINTENANCE

4.0 OPERATION ISSUES

- Monthly walk down check list to be made with maint. & operation for any defects for all towers.
- Quality of makeup water to OAC and OAC water is to be maintained as per design values.
- It should be checked that no water other than make up water from any other source is mixing with OAC.
- Cold water basin overflow is not there and level of OAC is being maintained accordingly.
- Chlorine dosing should be done continuously such that residual chlorine is maintained between 0.5 to 0.8 ppm.

5.0 PERFORMANCE CHECKS

5.1 The cooling tower performance check should be done exhaustively by EEMG every year taking all the readings like CW flow, Wet Bulb Temp, Cold Water Temp, Hot Water Temp. The procedure for this performance check will be as per ATC 105 for which CENPEEP has issued the guidelines in March 2005.

5.2 Cooling Tower effectiveness must be checked every month EEMG by using the formula given below, assuming the water flow as constant as taken in 5.1 above. The Targeted value of effectiveness should be more than 60% and as near the design value as possible.

$$\text{Effectiveness} = \frac{\text{Range}}{\text{Range} + \text{Approach}} \times 100\%$$

Range = Hot water temp. – Cold water temp.

Approach = Cold water temp. – Wet bulb temp.

GUIDELINES FOR COOLING TOWER MAINTENANCE

6. EXCEPTION REPORTING

The exceptions encountered during the inspection are to be reported as per the following format –Responsibility : MM/EMD/O&M-CIVIL/OPERATION

Description of exception	Area	Location of exception	Remarks

7. RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to preventive measures, inspection and testing of above systems as detailed in the LMI are with HODs of MM, O&M-Civil, Operation, EMD.

8. RECORD KEEPING

The records to be maintained under this LMI include:

- a) Compliance record of the activities detailed under this LMI
- b) Exception reporting, if any.

The above records are to be maintained by HODs of Operation, MM, O&M-CIVIL and EMD for the relevant responsibilities.

9. REVIEW

The document shall be reviewed by HOP, Simhadri on Two yearly basis or as necessary.

GUIDELINES FOR COOLING TOWER MAINTENANCE

REF NO. LMI/OGN/OPS/MECH/024/**DISTRIBUTION LIST**

Sl. No.	Date of issue	Issued to	Copy no.
1		GM(O&M)	
2		AGM(MM-I/C)	
3		AGM (C&I-I/C)	
4		AGM(EM-I/C)	
5		AGM(OPN)	
6		AGM(O&M-CIVIL)	
7		AGM(MTP)	
8		AGM(MM-OS)	
9		SCE	
10		AGM(EEMG)I/C	

GUIDELINES FOR COOLING TOWER MAINTENANCE

REF NO. LMI/OGN/OPS/MECH/024/**REVISION INDEX****(for changes/revision)**

Sl. No.	Rev. no.	Date	Details	Approved by
1	Rev:2, Issue-1	10-06-2022	No Change	

GUIDELINES FOR COOLING TOWER MAINTENANCE

COMPLIANCE RECORD		
REF NO. LMI/OGN/OPS/MECH/024/		
SYSTEM / EQUIPMENT DESCRIPTION	ACTIVITY	COMPLIANCE